

Personal Finance Management System

SUMMARY

Build a Personal Finance Management System as a modular web application for long-term personal money management.

PROJECTS

Personal Finance Management System | Modular web application for long-term personal money management.

- Understand → Financial Dashboard
- Track → Expense Recorder
- Share → Bill Splitter
- Plan → Financial Planner
- Commit → Instalment Tracker
- Reduce → Credit Card Payoff
- Grow → Savings & Investment
- Convert → Currency Converter
- Core design principle: Record once, analyze many times.
- Central transaction types: INCOME, EXPENSE, TRANSFER.
- All important user data must be persisted in a database with historical records available for searching, filtering, editing, reporting, comparison, trend analysis, and summaries.

CORE DESIGN PRINCIPLE

Record once, analyze many times.

PERSISTENT DATA REQUIREMENTS

Persist transactions, accounts, categories, sub-categories, bills, bill participants, bill settlements, budgets, savings goals, instalment plans, instalment payment history, credit cards, credit card payment history, savings contributions, investments, investment contributions/transactions, optional currency conversion records, and user settings. Include ID, Created Date, Updated Date, and status where applicable with proper relational relationships.

MODULES AND SCOPE

M1 Dashboard: KPIs, period selection, account overview, income vs expense, expense breakdown, spending trend, historical comparison. M2 Expense Recorder: income/expense/transfer entry, expense fields, history, search/sort/filter/edit/delete, daily/monthly/yearly summaries, category management, account/wallet management, balance logic. M3 Bill Splitter: bill creation, split methods, settlements, bill history, expense integration. M4 Financial Planner: budget management and savings goals. M5 Instalment Tracker: commitments, payment history, countdown. M6 Credit Card Payoff: card tracking, payoff calculator, snowball/avalanche strategies, payment history. M7 Savings & Investment: savings tracking/history and manual investment tracking. M8 Currency Converter: manual conversion, reverse conversion, buy/sell calculation, swap currency.

GLOBAL REQUIREMENTS

Multi-currency support across system with configurable base currency (default MYR), global search/filtering across records, long-term historical reporting, notifications/reminders, and strict data integrity rules (e.g., transfers are not income/expense; preserve original currency amounts; maintain historical consistency).

FINAL PRODUCT GOAL

A complete personal financial management platform following: Understand → Track → Share → Plan → Commit → Reduce → Grow → Convert. Prioritize accurate financial calculations, persistent historical data, clear reporting, modular architecture, and simple user experience.